

```

`define N 32
`define NOP 32'h00000013 // addi x0,x0,0
// `define SIM

// Instruction Memory
`define INSTR_MEM_SIZE 20'h1000 // in bytes
`define PC_RESET 32'h00000000

// Data Memory
`define DATA_MEM_SIZE 20'h1000 // in bytes
`define DATA_START 32'h80002000

// Opcodes
`define R 7'b0110011
`define I 7'b0010011
`define LD 7'b0000011
`define S 7'b0100011
`define B 7'b1100011
`define J 7'b1101111
`define JR 7'b1100111
`define U 7'b0110111
`define UPC 7'b0010111
`define CSR 7'b1110011
`define FENCE 7'b0001111
`define AMO 7'b0101111 // Opcode for Atomic Instructions

// ALU Control Signals
`define ADD 4'b0000
`define SUB 4'b0001
`define AND 4'b0010
`define OR 4'b0011
`define XOR 4'b0100
`define SRL 4'b0101
`define SLL 4'b0110
`define SRA 4'b0111
`define BUF 4'b1000
`define SLT 4'b1001
`define SLTU 4'b1010
`define EQ 4'b1011
`define GE 4'b1100
`define GEU 4'b1101

// Atomic Instructions
// func3 values for atomic operations
`define AMO_W 3'b010 // AMO operations on word (32-bit)
`define AMO_D 3'b011 // AMO operations on doubleword (64-bit, not used in 32-bit

```

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// func5 values for atomic operations
`define AMO_LR    5'b00010 // Load-Reserved
`define AMO_SC    5'b00011 // Store-Conditional
`define AMO_SWAP  5'b00001 // Atomic Swap
`define AMO_ADD   5'b00000 // Atomic Add
`define AMO_XOR   5'b00100 // Atomic XOR
`define AMO_AND   5'b01100 // Atomic AND
`define AMO_OR    5'b01000 // Atomic OR
`define AMO_MIN   5'b10000 // Atomic MIN (signed)
`define AMO_MAX   5'b10100 // Atomic MAX (signed)
`define AMO_MINU  5'b11000 // Atomic MIN (unsigned)
`define AMO_MAXU  5'b11100 // Atomic MAX (unsigned)
```

```
//func3 for Load Instructions
```

```
`define LB    3'b000
`define LH    3'b001
`define LW    3'b010
`define LBU   3'b100
`define LHU   3'b101
```

```
//func3 for Branch Instructions
```

```
`define BEQ   3'b000
`define BNE   3'b001
`define BLT   3'b100
`define BGE   3'b101
`define BLTU  3'b110
`define BGEU  3'b111
```

```
//func3 for Arithmetic and Logical Instructions
```

```
`define ADDI   3'b000
`define SUBI   3'b000
`define SLLI   3'b001
`define SLTI   3'b010
`define SLTUI  3'b011
`define XORI   3'b100
`define SRLI   3'b101
`define SRAI   3'b101
`define ORI    3'b110
`define ANDI   3'b111
```

```
// func3 for CSR Instructions
```

```
`define CSRRW  3'b001
`define CSRRS  3'b010
`define CSRRC  3'b011
```

```
`define CSRRWI 3'b101
`define CSRRSI 3'b110
`define CSRRCI 3'b111

// func7 for CSR-ECALL/EBREAK
`define ECALL 7'b0000000
`define EBREAK 7'b0000001

//// M-mode CSRs ////
// Machine-info
`define MVENDORID 12'hF11
`define MARCHID 12'hF12
`define MHARTID 12'hF14
// Machine Trap Setup
`define MSTATUS 12'h300
`define MISA 12'h301
`define MIE 12'h304
`define MTVEC 12'h305
// Machine Trap Handling
`define MSCRATCH 12'h340
`define MEPC 12'h341
`define MCAUSE 12'h342
`define MTVAL 12'h343
`define MIP 12'h344
```